

Adnan Harun Dogan

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EDUCATION

- 🎓 **Master of Science in Computer Engineering** Sep 2022 – Jan 2025
Middle East Technical University (METU) Ankara, Turkey
- **Thesis:** [Incorporating Piecewise Linear Functions with Constant Regions in Backpropagation](#)
 - **Advisor:** Prof. Dr. [Sinan Kalkan](#), Assoc. Prof. Dr. [Emre Akbas](#)
- 🎓 **Bachelor of Science in Computer Engineering** Sep 2018 – Jul 2022
Middle East Technical University (METU) Ankara, Turkey

EXPERIENCE

-  **Graduate Intern** (Supervisor: Prof. Dr. [Christian Holz](#)) Mar 2025 – Oct 2025
Sensing, Interaction, & Perception Laboratory ([SIPLab](#)), ETH Zürich Zürich, Switzerland
- Developed [learned frequency-weighted Kalman filters](#) that denoise the innovation spectrum under spectral supervision
 - Built [EEG multitaper-spectrum deep models](#) (ConvLSTM) for real-time VR cybersickness detection, +30% over baselines
-  **Research Assistant** (Supervisor: Prof. Dr. [Sinan Kalkan](#), Assoc. Prof. Dr. [Emre Akbas](#)) Feb 2023 – Present
Image Processing and Pattern Recognition Laboratory ([ImageLab](#)), METU Ankara, Turkey
- Contributed to [Bucketed Ranking Loss](#), speeding up ranking-based training of DETR/transformer detectors by up to 6x
 - Developed optimal-transport (Sinkhorn-Knopp) and ranking-based loss formulations for end-to-end object detection
- 🚀 **Undergraduate Intern** (Supervisor: Assoc. Prof. Dr. [Erol Sahin](#), Assoc. Prof. Dr. [Bugra Koku](#)) Oct 2021 – Feb 2022
Center for Artificial Intelligence and Robotics ([ROMER](#)), METU Ankara, Turkey
- Built a [self-driving stack](#) for a physical RC car, and trained RL agents for autonomous navigation task in ROS/Gym/PyTorch
 - Simulated a [Hector-quadrotor swarm](#) in ROS/Gazebo, approximating numerical solvers for multi-agent localization
-  **Undergraduate Intern** (Supervisor: Prof. Dr. [Mohammad Abdullah Al-Faruque](#)) Jun 2021 – Sep 2021
Cyber-Physical Systems Laboratory, University of California, Irvine Irvine, California, USA
- Built [multi-modal deep models](#) for beat-by-beat cardiac and sleep abnormality detection from physiological signals
 - Fused time-series and spectral features for arrhythmia classification, competitive in international challenges

PUBLICATIONS

- Dogan, A. H.**, Demirel, B. U., Holz, C. (2026). *Frequency-Weighted Neural Kalman Filters*. **ICRA 2026**. [PDF](#) · [Code](#)
- Demirel, B. U., **Dogan, A. H.**, Rossie, J., Möbus M., Holz, C. (2025). *Beyond subjectivity: Continuous cybersickness detection using eeg-based multitaper spectrum estimation*. **IEEE/TVCG 2025**. [DOI](#) · [Code](#)
- Yavuz, F., Cam B.C., **Dogan, A. H.**, Oksuz, K., Kalkan, S., Akbas, E. (2024). *Bucketed Ranking Loss for Efficient Ranking-based Training of Object Detectors*. **ECCV 2024**. [DOI](#) · [PDF](#) · [Code](#)
- Amin M. A., **Dogan, A. H.**, Kuru E. S., Sever Y., Angin P. (Apr 2024). *Misuse Detection and Response for Orchestrated Microservices Based Software*. **AINA 2024**. [PDF](#) · [Data](#)
- Karagoz, P., Cekinel, R. F., **Dogan, A. H.**, Oktay, B., Ozturk, A. U., Tonay, S. T., Tunel, B. M. (2024). *Enhancing Underground Built Heritage Analysis with Text Mining: A Case Study on Cappadocia*. Book Chapter 2024. [PDF](#)
- Sever Y., **Dogan, A. H.** (Aug 2023). *A Kubernetes dataset for misuse detection*. **ITU Journal 2023**. [DOI](#) · [PDF](#) · [Code](#)
- Sever, Y., Ekinici, G., **Dogan, A. H.**, Alparslan, B., Gurbuz, A. S., Jabrayilov, V., Angin, P. (Sep 2022). *An Empirical Analysis of IDS Approaches in Container Security*. **IEEE/SRMC 2022**. [🏆 Best Paper Award](#) [DOI](#)
- Demirel, B. U., **Dogan, A. H.**, Al-Faruque, M. A. (2021). *Two Might Do: A Beat-by-Beat Classification of Cardiac Abnormalities using Deep Learning and Domain-Specific Features*. **CinC 2021**. [DOI](#) · [PDF](#) · [Code](#)
- Buyukbas, E. B., **Dogan, A. H.**, Ozturk, A. U., Karagoz, P. (Sep 2021). *Explainability in Irony Detection*. **DaWaK 2021**. [PDF](#)
- Dogan, A. H.**, Dogan, A. (Jun 2021). *An Assembled Deep Learning Approach for Flow Field Prediction*. [PDF](#)

SCORES, HONORS, & AWARDS

- TOEFL iBT: 109/120 (R29 L29 S25 W26)**, ETS Oct 2022
- University Entrance Exam: ranked 1466th of 2.27M candidates**, OSYM, Turkey Aug 2017

TECHNICAL SKILLS & RESEARCH INTERESTS

Languages & Tools: 🐍 Python/PyTorch, 📀 C/C++, 📦 ROS2, 🦀 Rust

DevOps & HPC: 🐙 GitHub, 🐳 Docker, Slurm (Workload Manager), Apptainer

Differentiable Optimization: Combinatorial Optimization, Optimal Transport, Deep State Space Models

Deep Learning: Computer Vision, Self-Supervised Learning, Probabilistic Deep Learning, Deep Reinforcement Learning